



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 4 • STANDARD 6.AT.4

Determine the Whole Given the Part and Percent

Lesson 4-5 • Teacher Copy

Name: _____

Date: _____

Class: _____



TODAY'S OBJECTIVES

Content Objective: I can determine the whole when I know a part and the percent that part represents.

Language Objective: I can explain how I found the whole using the words part, whole, percent, and equation.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Determine the Whole Given the Part and Percent

Part

Whole

Double number line

Equation

Percent



Tap any word to see what it means and a picture.

1

Working backwards from a part and its percent finds the

2

How much of the base you get — your answer — is the

3

The full amount a percent is measured against is always

 %

4

Two parallel lines pairing amounts with their percents form a

5 The percent number sentence is part = × .

6 The word that means “per hundred” is .

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

What did the car cost when they bought it, and how does your method show it?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Determine the Whole Given the Part and Percent** — Working backwards from a part and its percent to find the whole amount it came from.

Hallar el total dados la parte y el porcentaje — Trabajar al revés a partir de una parte y su porcentaje para hallar el total del que proviene.

- ☐ **Part** — The amount you already know — the piece of the whole that the percent describes.

Parte — La cantidad que ya conoces: el pedazo del total que describe el porcentaje.

- ☐ **Whole** — The full amount the percent is measured against. The whole is always 100%.

Total — La cantidad completa contra la que se mide el porcentaje. El total siempre es 100 %.

- ☐ **Double number line** — Two lines lined up — dollars on one, percents on the other — so matching marks show the same amount two ways.

Recta numérica doble — Dos rectas alineadas — dólares en una y porcentajes en la otra— para que las marcas correspondientes muestren la misma cantidad de dos formas.

- ☐ **Equation** — A math sentence with an equal sign. Here it says percent $\times$ whole = part, with the whole unknown.

Ecuación — Una oración matemática con signo de igual. Aquí dice porcentaje $\times$ total = parte, con el total desconocido.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

The \$26,600 is the ____ and the purchase price is the _____. Written as an equation, $0.7v =$ _____. Dividing both sides by _____ gives $v =$ _____. The whole is _____ than the part, which makes sense because 70% is less than 100%.

Di tu respuesta primero: Mi respuesta es _____ porque _____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: A discount lowers the price, while tax, tip, and markup raise it.

Evidence: Students sort discount as lowering price and tax/tip/markup as raising it, and recognize all are a percent of a price.

Reasoning: This evidence connects the definition of determine the whole given the part and percent to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- The \$26,600 is the ____ and the purchase price is the _____. Written as an equation, $0.7v = \underline{\hspace{1cm}}$. Dividing both sides by ____ gives $v = \underline{\hspace{1cm}}$. The whole is ____ than the part, which makes sense because 70% is less than 100%.

- My answer is ____.

Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- My claim is ____.

Mi afirmación es ____.

- I know because ____.

Lo sé porque ____.

- So ____.

Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
- **My math evidence is ____.**
Mi evidencia matemática es ____.
- **This proves ____ because ____.**
Esto demuestra ____ porque ____.

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.
- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.
- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Determine the Whole Given the Part and Percent
2. **Notes 2:** Part
3. **Notes 3:** Whole
4. **Notes 4:** Double number line
5. **Notes 5:** Equation
6. **Notes 6:** Percent
7. **Watch for this mistake:** *A common mistake when finding the whole is multiplying by the percent instead of dividing. For "\$42 is 60% of the original price," computing $0.6 \times 42 = \$25.20$ makes the original price SMALLER than the sale price — impossible for a discount. Since $\text{percent} \times \text{whole} = \text{part}$, the whole is found by dividing: $42 \div 0.6 = \$70$. Whenever the percent is less than 100%, the whole must come out larger than the part, so a smaller answer is the tell.*
8. **Try It 1:** — *"Of what number?" signals a missing whole. "What is ___% of ___?" hands you the whole and asks for the part.*
9. **Try It 2:** — *Every one is the same move: divide the part by the percent written as a decimal.*